Research for bachelor's degree

Proposed direction: extend PiCam (Raspberry Pi 5) with a secure two-way push-to-talk audio channel between the caregiver app and the device. I plan to compare three audio transports (record-and-play via Firebase, chunked streaming, WebRTC audio) on mouth-to-ear latency, jitter/packet loss and intelligibility, and to add authentication and telemetry, fall detection with voice check-in.

https://www.youtube.com/watch?v=BUEbC3vQNcI – Waveshare Pan-Tilt

https://github.com/mazupo/pi-webrtc - Low-latency hardware WebRTC video streaming and remote control for Raspberry Pi

https://ieeexplore.ieee.org/document/9352144 - YOLO Based Real-Time Human Detection for Smart Video Surveillance at the Edge

https://www.mdpi.com/1999-5903/17/8/324 - Real-Time Fall Monitoring for Seniors via YOLO and Voice Interaction

https://research.ustp.at/en/projects/smart-companion-2 Smart Companion 2

https://ieeexplore.ieee.org/document/7868363 - Fall detection system for the elderly

https://indico.freedesktop.org/event/11/contributions/466/attachments/341/443/CuttingAudioLatency.pdf - Cutting audio latency with bidirectional WebRTC

New components for my projects:

- ➔ BME280 or SHT4x: Replaces DHT22, for more precise result and avoids driver problems on Pi 5
- ➔ Official 27 W PSU + active cooler: Stable power and cooling
- ➔ Physical buttons**:** "I'm OK" / PTT, plus a status LED and buzzer on GPIO: Non-voice confirmation and a local alarm
- ➔ USB speakerphone (mic + speaker in one): Voice prompt, PTT, spoken response
- ➔ INMP441 mic + MAX98357A amp + 3 W speaker(alternative for USB speakerphone)
- ➔ UPS HAT / battery: Survive power cuts